

Number: 02

Title: 3D human pose estimation in video with temporal convolutions and semi-supervised training

Link: <https://arxiv.org/pdf/1811.11742.pdf>

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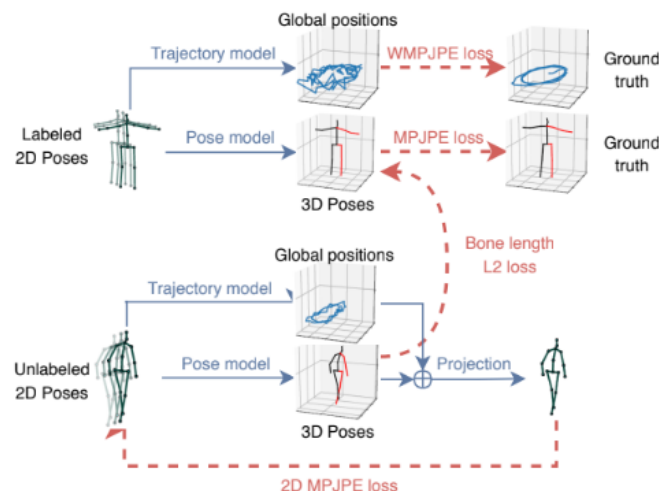
Estimates 3D poses with fully convolutional models based on 2D keypoints. It introduces back-projection, a semi-supervised approach only requiring camera intrinsic parameters.

Basic principles:

- Two-step pose estimation
- Video pose estimation. Temporal information
- Semi-supervised

Method:

It works as a semi-supervised approach combining labeled and unlabeled data.



- Predict 2D keypoints for unlabeled video - Estimate 3D poses - Back-project
- Global position is needed. Otherwise, the back-projection to 2D can project at the center of the screen with a fixed scale. Therefore the 3D trajectory must be regressed too. Two networks with the same architecture but share no weight
- For getting plausible 3D poses add soft constraints to approximately match the mean bone lengths of the subjects in the unlabeled batch to the subjects of the labeled batch. Good for self-supervision

Evaluation: Human3.6M & HumanEva-I

- **Protocol 1:** Mean per-joint position error (MPJPE) in millimeters which is the euclidean distance between predicted joints positions and ground-truth joints positions
- **Protocol 2:** Reports error after alignment with the ground truth in translation, rotation and scale (P-MPJPE)
- **Protocol 3:** Aligns predicted poses with the ground-truth only in scale (N-MPJPE)

WMPJPE: Weighted MPJPE. It is harder to regress precise trajectory if the subject is further away from the camera.